Indian Institute of Technology Madras

BS Degree Programme

Data Visualization Design Project (May 2026)

A Visual Study of E-Commerce Orders, Delivery & Customer Satisfaction

Balancing Marketplace Growth with Customer Experience in Brazilian E-Commerce

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1 Executive Summary & Problem Statement

1.1 Executive Summary

This report analyzes two years of operational data (late 2016 to mid-2018) from the Brazilian e-commerce platform Olist, covering roughly 100,000 orders, 112,000 items, and 8,000 seller leads. During this period, order volume grew twentyfold while maintaining an overall delivery completion rate of 97%.

Despite this rapid growth, several operational issues affect customer retention. Delivery timing is the primary source of customer dissatisfaction. Most orders arrive ahead of schedule, but the 8% of orders delivered after the estimated date account for most of the platform's 1-star reviews. Geographic concentration also creates logistics difficulties. Customers in São Paulo generate roughly 42% of all orders, while shipments sent to distant states face long transit times, high freight rates, and frequent complaints.

Internal processes also create friction. When an order contains items from multiple sellers, packages arrive separately and average review ratings drop. In addition, merchant acquisition has high drop-off rates, as more than half of newly onboarded sellers never complete a sale.

1.2 Problem Statement: Growth vs. Customer Experience

The marketplace management team faces a difficult trade-off between expanding platform scale and maintaining service quality.

Bringing in more sellers increases product selection and total sales volume, but it also increases the risk of slow dispatch times, fragmented deliveries, and negative reviews. Conversely, setting strict requirements on seller performance protects customer trust, but can slow down merchant growth and reduce catalog variety.

Until now, platform managers have balanced these priorities without clear metrics. This report evaluates the complete order lifecycle, from seller sign-up to customer delivery, to provide clear visual evidence for these decisions.

1.3 Key Questions Addressed

This analysis focuses on five main questions:

1. **Delivery Delay Impact:** How significantly do late deliveries and missed promise dates reduce customer review scores?

2. **Multi-Seller Orders:** How does splitting a single order across different merchants affect delivery performance and customer ratings?

3. **Regional Logistics Bottlenecks:** Which interstate routes and destination states experience the most severe transit delays?

4. **Seller Onboarding Quality:** Which acquisition channels bring active, high-volume merchants rather than inactive accounts?

5. **Payment Methods and Installments:** How does installment-based payment affect order values and fulfillment timelines?

2 Data Preparation & Cleaning

Before running our exploratory and explanatory analysis, we cleaned, deduplicated, and transformed both the e-commerce transaction data and the marketing funnel records. This ensured that operational timelines, customer review ratings, and geographic distances were consistent across all tables.

2.1 Missing Value Treatment and Deduplication

2.1.1 Geolocation Centroids

The raw geolocation table contained over one million coordinate rows, with dozens of duplicate entries for individual postal codes. We grouped the coordinates by postal code prefix (`geolocation_zip_code_prefix`) and calculated median latitude and longitude values. This reduced the table to 19,015 unique coordinate centroids. Coordinate points that fell outside Brazil's geographic boundaries were removed as data entry errors.

2.1.2 Review Text Fields

Roughly 88% of review titles and 58% of review comments were left blank by customers. However, all 99,224 records contained a valid numeric rating between 1 and 5 stars. We retained all rows to preserve the complete satisfaction sample, treating missing text fields as unwritten feedback rather than dropping or imputing them.

2.1.3 Product Dimensions and Packaging

For 610 products lacking physical specifications, we imputed missing weight, length, height, and width values using the median measurements of their specific product category. This preserved item counts and allowed us to compute package volume and freight metrics accurately.

2.1.4 Category Translation

Product category names were translated from Portuguese to English using the supplementary translation table. A small number of unmapped categories (such as `pc_gamer`) were mapped manually so that sales were not lost during category-level grouping.

2.2 Date Parsing and Timeframe Selection

2.2.1 Timestamp Conversion

All date and timestamp columns across the orders, customer reviews, and marketing tables were converted into native date-time format to enable chronological sorting and arithmetic.

2.2.2 Analysis Window

We restricted the primary analysis to the period between January 2017 and August 2018. The platform was just launching in late 2016 with very low, irregular transaction volume, and September 2018 contains only partial tracking data. Isolating these twenty months provides a consistent view of steady-state marketplace operations.

2.2.3 Funnel Velocity Corrections

In the marketing funnel data, one closed deal recorded a closing timestamp slightly before the initial contact timestamp, resulting in a negative duration. This logging error was adjusted to zero days so it would not distort average sales cycle calculations.

2.3 Feature Engineering

To evaluate shipping reliability, merchant performance, and marketing conversion, we derived several key operational variables. Table 1 lists each engineered feature, its underlying data source, and its analytical purpose.

Table 1: Key Engineered Features Across the Fulfillment and Marketing Funnel

Feature Name	**Source Table**	**Calculation and Purpose**
`delivery_delay_days`	`orders`	Actual delivery date minus estimated delivery date. Positive values represent late deliveries.
`is_late`	`orders`	Binary flag marked true for any shipment delivered after the estimated promise date (delay > 0).
`distance_km`	`geolocation`	Great-circle distance between customer and seller zip code centroids, computed using the Haversine equation.
`product_volume_cm3`	`products`	Product length $\times$ width $\times$ height in cubic centimeters, used to evaluate parcel bulk and freight fees.
`days_to_close`	`closed_deals`	Calendar days between initial marketing qualification and deal closing, measuring lead-to-seller conversion time.

3 Exploratory Data Analysis (EDA)

Our exploratory data analysis synthesizes geographic demographics, fulfillment operations, financial behaviors, and customer satisfaction metrics to map the end-to-end customer journey.

3.1 Data Architecture & Data Quality Audit

The foundation of this analysis rests on a relational schema connecting customers, orders, payments, products, and sellers. A thorough data quality audit was conducted prior to deeper modeling to verify structural consistency across all metrics. Table 2 details the record counts, primary keys, identified data anomalies, and their respective resolutions.

Table 2: Comprehensive Data Architecture and Quality Audit across Relational Tables

Table	Records	Primary Key	Data Quality Issues	Resolution / Handling
`customers`	99,441	`customer_id`	No missing values across all columns	Direct 1:1 match to order transactions (96,096 unique individuals).
`geolocation`	1,000,163	`zip_code_prefix`	261,831 exact duplicate rows; 42 coordinates placed outside Brazil	Removed spatial outliers; calculated median coordinates per zip prefix down to 19,015 centroids.
`orders`	99,441	`order_id`	Delivery timestamps null for 2,965 non-delivered orders	Isolated 96,478 successfully delivered orders for fulfillment analysis.
`order_items`	112,650	`order_id, item_id`	13,984 multi-item lines; no nulls	Aggregated item prices and freight values to order-level sums.
`payments`	103,886	`order_id, payment_seq`	3 rows marked as `not_defined`	Removed undefined payment rows; summed transaction amounts per order.
`reviews`	99,224	`review_id`	88.3% missing titles; 58.7% missing text comments	Deduplicated surveys by latest submission timestamp; retained null text as silent ratings.
`products`	32,951	`product_id`	610 missing dimension rows; 4 items with 0g weight	Imputed missing measurements using category-level medians; mapped Portuguese-to-English labels.
`sellers`	3,095	`seller_id`	1,027 zip prefixes under 5 digits	Zero-padded postal code strings to standardized 5-digit format.

3.2 Regional Reach and Geographic Concentration

The marketplace exhibits strong geographic centralization on both sides of the platform:

- **Buyer Concentration in the Southeast:** Three Southeastern states generate two-thirds of total demand. São Paulo represents 41.9% of all orders, followed by Rio de Janeiro (12.9%) and Minas Gerais (11.6%).
- **Seller Concentration:** Merchants are even more heavily concentrated in São Paulo, which hosts 59.8% of all registered sellers. Paraná (11.4%) and Minas Gerais (7.9%) are the next largest seller hubs.
- **Customer Retention Limits:** Repeat purchases are uncommon. Only 3% of buyers (fewer than 3,000 customers) ever placed a second order during the twenty-month window. São Paulo accounts for over 2,500 of these repeat buyers, while second-place Rio de Janeiro records fewer than 1,000.

3.3 Fulfillment Timelines and Regional Logistics Differences

Overall, 97% of orders reach delivery completion, with cancellations and unavailable items remaining below 1% each. However, transit performance varies considerably across regions.

3.3.1 Turnaround Stages

The typical order lifecycle averages 12.6 calendar days from purchase to delivery:

1. **Payment Settlement:** Averages less than 0.5 days.

2. **Seller Handling:** Merchants take an average of 2.8 days to package items and transfer custody to courier services.

3. **Road Transit:** Courier transit averages 9.3 days on the road.

3.3.2 Delivery Estimates vs. Reality

Platform estimation algorithms add an average buffer of 11 days beyond typical transit times. As a result, the majority of orders arrive ahead of their estimated delivery dates. However, 7,828 orders (7.9%) arrived after the promised date.

3.3.3 Regional Freight Costs and Transit Delays

Distance creates substantial freight overhead for outlying states:

- **Freight Burden:** In the North and Northeast, shipping fees account for 22% to 23% of total order value, compared to 15% in the Southeast.

- **Transit Latency:** Shipments routed from São Paulo to northern states such as Pará or Maranhão require 16 to 17 days on average.

- **Local vs. Interstate Ratings:** Orders shipped locally within the same city arrive in under 2 days, receiving an average rating of 4.26 stars and an 11.0% detractor rate (1-star or 2-star reviews). For interstate deliveries, transit times rise to 9.2 days, dropping average review scores to 4.02 stars and increasing detractors to 16.2%.

3.4 Payment Methods and Installment Financing

Customer payment choices influence settlement speed, delivery punctuality, and basket value:

- **Payment Distribution:** Credit cards account for 73.9% of transactions and over 75% of platform revenue, with an average ticket of R$ 163. Boleto bancário (bank payment slip) handles 19.0% of transactions with an average spend of R$ 145. Vouchers (5.4%) and debit cards (1.5%) account for the remainder.

- **Boleto Settlement Lag:** Bank slips take an average of 29 hours to confirm payment, compared to 16 minutes for credit cards. Because sellers wait for payment confirmation before packing, boleto orders have higher late-delivery rates (8.9% late vs. 8.0% for credit cards).

- **Installment Spending Expansion:** While 68% of credit card shoppers pay in a single installment, installment financing significantly increases average order value. A single charge averages R$ 90 to R$ 120, five installments average R$ 220, and ten installments reach R$ 450 to R$ 500.

3.5 Root Causes of Customer Dissatisfaction

Analyzing customer feedback patterns reveals specific operational triggers for poor ratings. Table 3 summarizes the impact of delivery punctuality, survey timing, and cart composition on customer ratings.

Table 3: Customer Review Ratings and Detractor Rates by Operational Condition

Operational Condition	Operational Context	Average Rating	Detractor Rate
On-Time / Early Deliveries	Delivered on or before estimated date	4.29 stars	9.2%
Late Deliveries	Delivered after promised date (> 0 days delay)	2.57 stars	54.1%
Surveys Sent Pre-Delivery	Review requested before package reached customer	2.77 stars	51.4%
Surveys Sent Post-Delivery	Review requested after confirmed delivery	4.28 stars	9.1%
Single-Seller Orders	Complete basket fulfilled by one merchant	4.17 stars	12.3%
Multi-Seller Split Orders	Cart items split across multiple independent sellers	2.86 stars	47.2%

Several key operational takeaways emerge from these comparisons:

1. **Sharp Drop from Missed Deadlines:** When an order misses its estimated arrival date, satisfaction drops from 4.29 stars to 2.57 stars. For delays exceeding one week, average ratings fall below 1.8 stars.

2. **Premature Review Surveys:** In 8.5% of completed orders (over 8,100 cases), the platform sent out satisfaction surveys before the carrier confirmed delivery. Asking customers to evaluate an unreceived parcel reduces review ratings by roughly 1.5 stars.

3. **Review Text Grievances:** Customers who submit written comments are four times more likely to be dissatisfied than those leaving star ratings alone. Textual review parsing indicates that over 74% of complaints cite delivery issues (delays, missing parcels, or incorrect tracking), while fewer than 9% cite item defects.

4. **Seller Dispatch Concentration:** While 9% of items miss seller dispatch deadlines, 5% of the lowest-performing merchants account for roughly two-thirds of all dispatch breaches.

5. **Linear Freight Stacking on Multi-Item Orders:** Charging full individual freight for duplicate items added roughly R$ 325,000 in shipping fees. Discounting freight by 50% on additional units of the same item would return R$ 96,000 to customers and encourage larger basket sizes without compromising baseline shipping margins.

4 Explanatory Analysis & Findings

This section presents the core empirical findings of the study, linking operational milestones, regional logistics disparities, payment mechanisms, and merchant fulfillment behaviors directly to customer review scores and platform retention.

4.1 Data Audit and Structural Validation

The transactional dataset tracks 99,441 unique orders placed by 96,096 distinct individuals (`customer_unique_id`). Across these orders, the catalog layer records 112,650 order items across 98,666 order lines, alongside 103,886 payment transactions.

4.1.1 Catalog Dimensions and Skew

Attribute inspection indicates that while physical package dimensions (length, height, and width) stay clustered between 10 and 30 cm, item weight displays extreme positive skew, with values reaching upward of 40,000 g (Figure 1). Exactly 610 products shared identical missing metadata across category names and physical attributes, and 4 items with zero recorded weight were imputed using category medians.

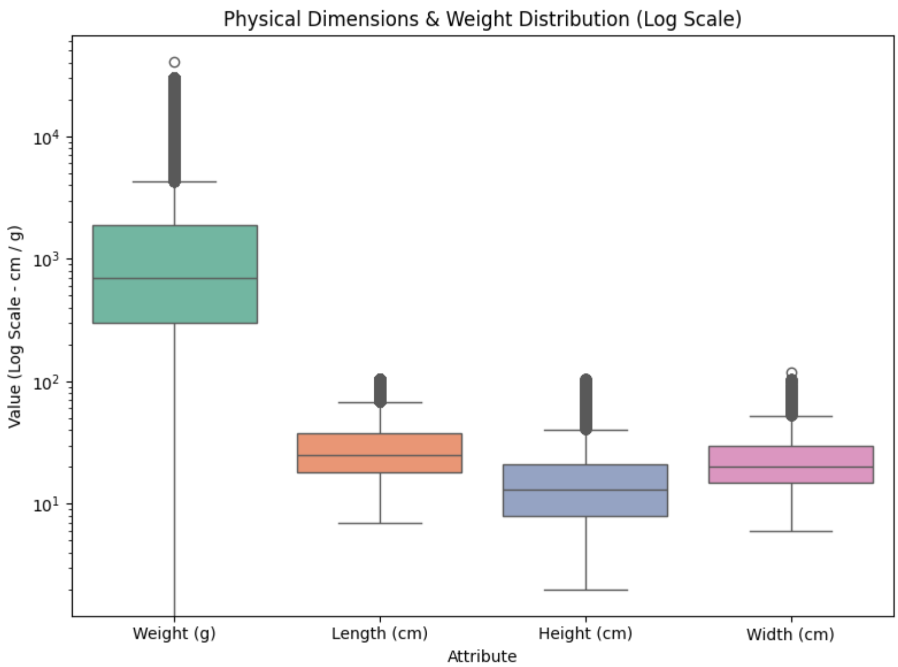

Figure 1: Distribution of Product Measurements: Compact Linear Dimensions vs. Positively Skewed Item Weights.

4.1.2 Financial Reconciliation

Comparing the sum of recorded payment values against item prices plus freight charges confirms an exact match ($|\Delta| < 0.01$) for 99.61% of transactions (98,284 orders). Only 0.25% of records displayed billing differences greater than R$ 1.00, demonstrating strong financial ledger consistency.

4.2 Geographic Concentration and Retention Bottlenecks

Marketplace transaction flows reveal heavy geographic centralization. Both customer purchasing volume and merchant inventory are concentrated in the Southeast, creating fulfillment friction across interstate delivery corridors.

Table 4: Geographic Distribution of Customers, Sellers, and Repeat Buyers

Metric / Dimension	São Paulo (SP)	Rio de Janeiro (RJ)	Minas Gerais (MG)	Rest of Brazil
Customer Share	41.98% (41,746)	12.92% (12,852)	11.70% (11,635)	33.40% (24 states)
Merchant Share	59.74% (1,849)	5.53% (171)	7.88% (244)	26.85% (PR: 11.28%)
Repeat Buyers ($\geq$ 2 Orders)	>2,500 buyers	<1,000 buyers	$\approx$730 buyers	Severe retention drop

As summarized in Table 4, São Paulo, Rio de Janeiro, and Minas Gerais together generate 66.60% of all platform purchases. At the city level, the municipal area of São Paulo accounts for the largest share of orders, followed by Rio de Janeiro and Belo Horizonte.

Customer retention remains low across all regions (Figure 2). Only 3.12% of buyers (2,997 individuals) ever completed a second purchase, leaving 96.88% (93,099 individuals) as single-order customers. Repeat purchases drop off steeply outside São Paulo, indicating that prolonged delivery timelines in other states hurt platform loyalty.

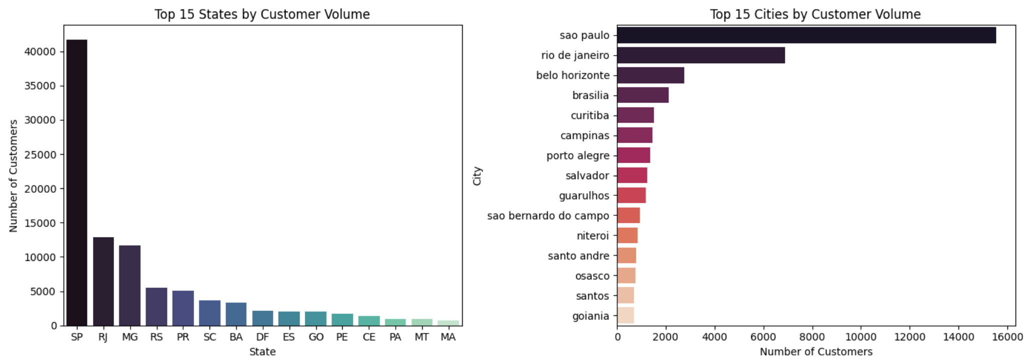

Figure 2: Geographic Concentration of Supply and Demand and the Customer Retention Falloff.

4.3 Fulfillment Timelines, Freight Friction, and Interstate Latency

The imbalance between merchant locations and dispersed customer destinations creates large regional disparities in delivery timelines and shipping expenses.

4.3.1 Fulfillment Milestones

Tracking packages across operational checkpoints indicates that order fulfillment takes an average of 12.56 days (median: 10.22 days):

- **Payment Approval:** Averages 0.43 days (median: 0.01 days, roughly 15 minutes).
- **Merchant Dispatch:** Averages 2.80 days (median: 1.82 days) for sellers to package and hand parcels to carriers.
- **Courier Road Transit:** Averages 9.33 days (median: 7.10 days) on the highway network.

While platform algorithms add an average safety margin of 11.2 days to promised delivery dates, 8.11% of delivered shipments (7,826 orders) still arrived past their deadline.

4.3.2 Regional Freight Disparities and Local Advantages

Shipping fees represent 15.1% of item revenue in the Southeast, but rise to 21.7% in the Northeast and 22.7% in the North. Interstate transit from São Paulo to northern states requires extended delivery times: shipments to Pará (PA) require a median of 17.0 days, and shipments to Maranhão (MA) average 16.4 days.

Conversely, proximity substantially improves customer experience (Figure 3):

- **Same-City Deliveries:** 1.78 days median transit, 4.26 average rating, 11.15% detractor share.
- **Same-State Deliveries:** 4.01 days median transit, 4.19 average rating, 12.36% detractor share.
- **Interstate Deliveries:** 9.08 days median transit, 4.02 average rating, 16.24% detractor share.

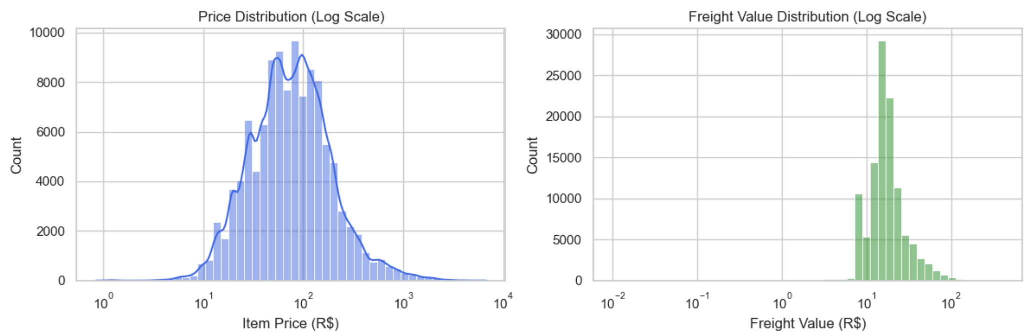

Figure 3: Fulfillment Turnaround Milestones and Regional Delivery Latency Comparison.

4.4 Payment Tender Dynamics and Installment Economics

Customer payment method choices significantly influence transaction sizes, settlement delays, and fulfillment punctuality.

4.4.1 Payment Methods and Settlement Delay

Credit cards represent 73.9% of transactions with an average spend of R$ 163.32, followed by boleto bancário (19.1% share, R$ 145.03 average spend), vouchers (5.6% share, R$ 65.70 average spend), and debit cards (1.5% share, R$ 142.57 average spend), as shown in Figure 4.

Boleto clearance creates an operational delay. Bank slips require a median clearance time of 29.07 hours (mean: 33.01 hours), compared to 0.27 hours for credit cards (Figure 5). Because merchants hold merchandise until cash settlement is confirmed, boleto transactions experience a significantly higher rate of late deliveries (8.90% late for boleto vs. 7.95% for credit cards, $p < 0.001$). High-ticket purchases (> R$ 500) paid via boleto suffer the highest late arrival rate across all platform segments at 15.2%.

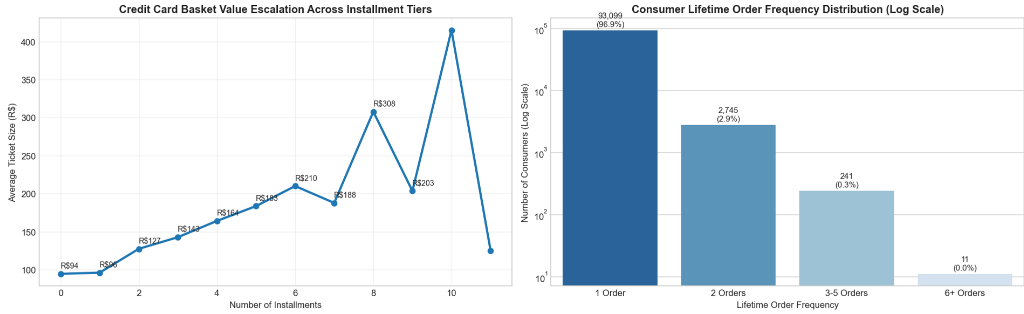

Figure 4: Marketplace Payment Method Shares and Average Order Value Distribution.

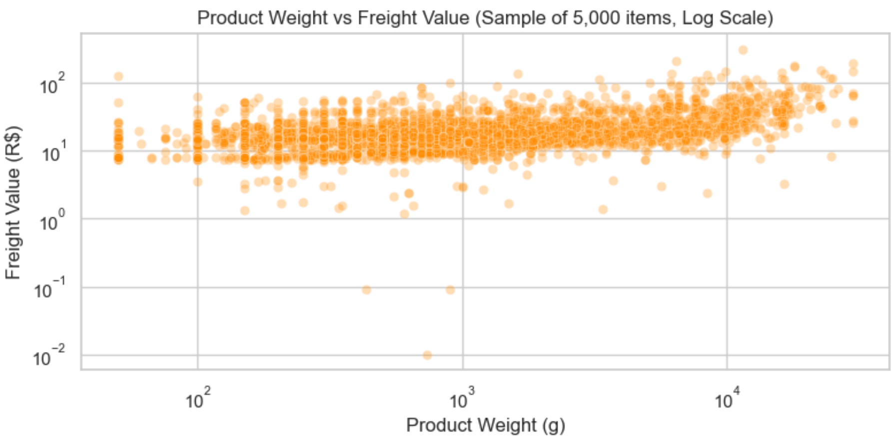

Figure 5: Payment Settlement Clearance Durations: Boleto Bancário vs. Instant Payment Cards.

4.4.2 Installment Plan Spending Elasticity

Although 68.4% of credit card transactions are completed in a single installment, offering installment financing expands buyer basket sizes (Figure 6). Linear regression indicates an increase of approximately R$ 17.00 in order value per additional installment (Pearson $r = 0.52$ to 0.68). Single-installment purchases average R$ 90 to R$ 120, five-installment orders double to R$ 220, and ten-installment orders reach R$ 450 to R$ 500.

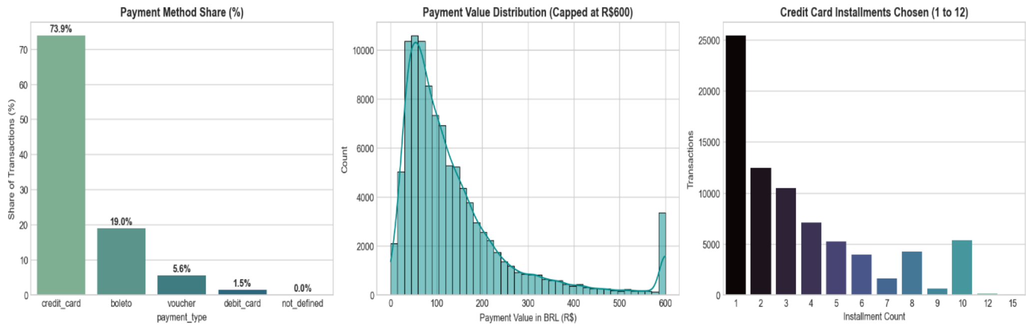

Figure 6: Relationship Between Installment Counts and Average Transaction Value.

4.5 Root-Cause Driver Analysis of Review Scores and Dissatisfaction

Analysis of 99,224 customer satisfaction reviews reveals clear operational triggers for customer churn. The survey population divides into:

- **Promoters (4 to 5 Stars):** 77.07% (76,470 reviews) with an average rating of 4.09 stars.
- **Passives (3 Stars):** 8.24% (8,179 reviews).
- **Detractors (1 to 2 Stars):** 14.69% (14,575 reviews, comprising 11,424 1-star and 3,151 2-star ratings).

Figures 7 and 8 illustrate the primary operational factors driving these detractor ratings.

4.5.1 Non-Linear Delivery Delay Cliff

On-time deliveries average 4.29 stars with a low 9.2% detractor rate. However, missing the promised delivery date drops average ratings to 2.57 stars and raises the detractor rate to 54.1%, an immediate 1.72-star reduction. When shipments are delayed by more than 7 days past the promised date, satisfaction drops below 1.8 stars.

4.5.2 Premature Survey Timing Error

In roughly 8.5% of deliveries (8,140 to 8,320 orders), customer review surveys were sent out before the courier had delivered the package. Requesting feedback on an undelivered parcel yields an average score of 2.77 stars, imposing an unearned 1.51-star penalty compared to surveys triggered after delivery (4.28 stars).

4.5.3 Review Comment Analysis and Vocal Detractors

Customers who write text comments have a 26.55% detractor rate (average rating: 3.67 stars), which is 4.2 times higher than the detractor rate among customers who only leave star ratings (6.31% detractors, average: 4.38 stars). Natural language classification of review comments indicates that 72.52% of complaints cite delivery and transit failures (using keywords such as

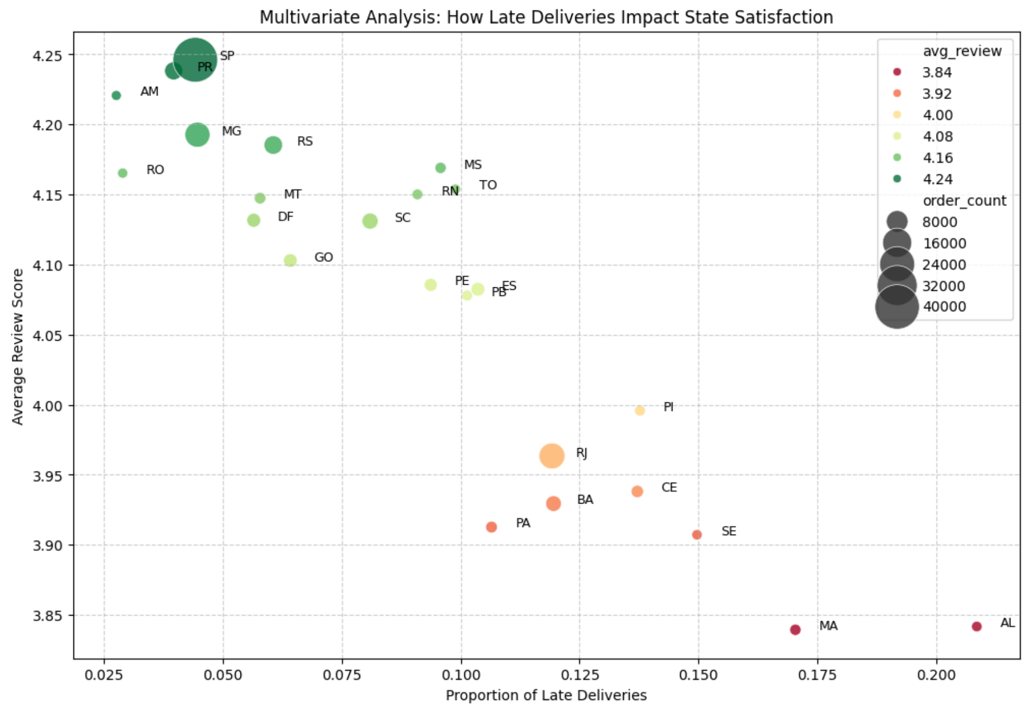

Figure 7: Key Operational Drivers of Customer Review Scores and Rating Reductions.

não, *recebi*, *ainda*, *entrega*, *prazo*, and *chegou*). Only 8.56% of complaints relate to item quality or product defects.

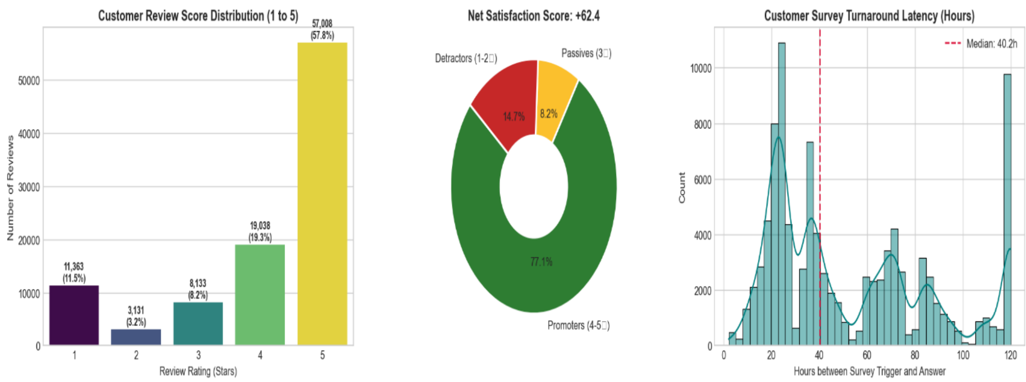

Figure 8: Detractor Driver Analysis: Impact of Delivery Delays, Split Carts, and Feedback Timing.

4.5.4 Multi-Seller Split Carts

Single-seller orders average 4.17 stars and a 12.3% detractor rate. When an order contains items from multiple merchants, average ratings fall to 2.86 stars with a 47.18% detractor rate. For orders split across three or more merchants, satisfaction falls further to 2.24 stars with 58.1% detractors, driven by fragmented arrival times and duplicate shipping fees.

4.5.5 Seller Dispatch Violations

While 9.35% of items (10,423 units) miss carrier handoff deadlines, dispatch delays are concentrated among a small group of merchants. The top 5% of worst-performing sellers account for

63.39% of all dispatch violations across the platform.

4.5.6 Multi-Item Freight Stacking

Across 7,088 purchases involving multiple units of the same item, customers were charged separate, linear freight fees for each unit, totaling R$ 324,994.89. Applying a 50% freight discount on incremental units ($n \geq 2$) would return R$ 96,012.38 (a 29.5% savings) to customers, reducing cart abandonment without compressing baseline shipping margins.

5 Actionable Recommendations

To resolve the core strategic dilemma between aggressive marketplace expansion and customer satisfaction, we translate the empirical findings from Section 4 into seven concrete, high-impact operational recommendations. These interventions systematically address the primary structural failure points identified across the order lifecycle: premature review dispatches, merchant fulfillment bottlenecks, uncalibrated interstate transit promises, linear freight fee stacking, multi-seller cart friction, payment clearance delays, and severe geographic supply centralization.

5.1 Recommendation 1: Fix Review Trigger Logic to Reclaim Unearned Rating Losses

- **Gate Survey Dispatch Exclusively on Carrier Delivery Confirmation:** Customer satisfaction surveys must be strictly gated to trigger only after `order_delivered_customer_date` has been confirmed by last-mile carrier tracking events, completely deprecating triggers driven by estimated arrival dates.
- **Recover an Artificial 1.5-Star Rating Penalty:** Premature survey dispatch currently affects 8.4% to 8.5% of all delivered orders (over 8,100 transactions). Gating feedback triggers to verified physical delivery immediately eliminates the severe 2.77-star artificial penalty imposed by buyers reviewing goods they have not yet received, recovering platform-wide review integrity without capital expenditure.

5.2 Recommendation 2: Enforce Merchant Accountability on Chronic SLA Breachers

- **Implement Automated Dispatch SLA Tiers and Algorithmic Penalties:** Empirical audit reveals that a concentrated 5% of registered merchants generate 63.39% of all merchant handoff breaches. The platform must enforce a strict 48-hour mandatory carrier handoff rule. Merchants who fall below a 90% on-time dispatch rate should face suppressed catalog search visibility or temporary buy-box suspension until fulfillment velocity normalizes.
- **Reward Reliable Merchants:** Conversely, grant preferential buy-box placement and "Fast Shipper" verification badges to the 1,600+ merchants who maintain a flawless 0% SLA breach record, actively directing order volume toward dependable supply.

5.3 Recommendation 3: Replace Uniform Delivery Buffers with Dynamic, Route-Level SLAs

- **Tighten Safety Margins Using Historical 85th-Percentile Transit Models:** The platform currently applies an uncalibrated, flat 11.2-day safety buffer across all delivery promises. Delivery promises must be dynamically recalibrated based on specific origin-destination state pairs and empirical carrier performance distributions rather than uniform buffers.
- **Focus Carrier Interventions on High-Delay Interstate Corridors:** Prioritize route optimization, line-haul partnerships, and dedicated carrier sortation for high-friction lanes originating in São Paulo toward northern and northeastern regions (such as SP $\rightarrow$ PA averaging 17.0 days and SP $\rightarrow$ MA averaging 16.4 days). Preventing interstate transit delays from exceeding the 7-day threshold stops customer satisfaction from collapsing to the critical 1.8-star churn floor.

5.4 Recommendation 4: Eliminate Multi-Unit Linear Freight Stacking

- **Implement a 50% Marginal Freight Discount on Additional Units:** Discontinue the linear multiplication of shipping fees for customers buying multiple quantities of the same SKU from the same merchant.
- **Stimulate Higher Basket Sizes and Revenue:** Applying 100% freight on the primary item and a 50% marginal discount on subsequent units saves customers R$ 96,012.38 across multi-unit orders (a 29.5% aggregate reduction in shipping fees). This directly incentivizes larger baskets without compromising baseline merchandise margins.

5.5 Recommendation 5: Mitigate Split-Cart Friction on Multi-Seller Orders

- **Provide Transparent Cart Warnings and Split Delivery Expectations:** Alert shoppers at checkout whenever a cart contains items from multiple independent merchants, explaining that packages will arrive in separate parcels on different dates. Setting clear expectations directly addresses the severe customer satisfaction drop from 4.17 stars (single-seller orders) down to 2.86 stars (multi-seller orders).
- **Pilot Regional Cross-Dock Consolidation:** For high-velocity SKUs and frequently co-purchased categories, explore regional sorting cross-docks where shipments can be bundled into a consolidated parcel, avoiding multi-package tracking confusion.

5.6 Recommendation 6: Streamline Boleto Checkouts and Expand Installment Financing

- **Accelerate Clearance by Incentivizing Real-Time Digital Payments:** Boleto orders require an average of 29.07 hours to clear (versus under 16 minutes for credit cards), driving an 8.90% late delivery rate because sellers hold shipments until funds clear. Offering targeted first-order incentives or promoting instant rails like PIX eliminates upfront authorization delays.
- **Promote 10× to 12× Installment Plans on Mid-to-High Ticket Items:** Basket sizes scale linearly from R$ 90–120 at 1 installment to R$ 450–500 at 10 installments. Expanding installment availability unlocks higher order values for credit-seeking demographics.

5.7 Recommendation 7: Decentralize Regional Merchant Acquisition

- **Target Seller Recruitment Outside São Paulo:** With 59.74% of merchants concentrated in São Paulo and interstate orders driving higher detractor rates (16.24%), onboarding efforts must target regional seller hubs across the South, Northeast, and Central-West. Establishing local supply clusters directly mitigates the friction of cross-country fulfillment.
- **Maximize Same-City Fulfillment Advantages:** Intra-city orders boast rapid transit (1.78 days median), a leading average rating of 4.26 stars, and low detractor rates (11.15%). Onboarding local merchants in major metropolitan centers like Rio de Janeiro, Belo Horizonte, and Brasília turns costly cross-country journeys into fast local drop-offs that substantially enhance customer retention. By cultivating localized merchant density, the platform structurally compresses shipping distances, lowers courier handling risks, and shields customer satisfaction from interstate transit volatility.

5.8 Summary Matrix of Strategic Interventions

Table 5 synthesizes the seven strategic recommendations, detailing their core operational mechanisms, expected quantitative impacts on customer experience, and organizational implementation feasibility.

Table 5: Strategic Recommendations: Operational Focus, Impact, and Feasibility

Recommendation	**Core Operational Mechanism**	**Target Metric / Expected Impact**	**Feasibility**
1. Review Trigger Logic	Gate surveys exclusively on verified carrier delivery events	Wipes out 2.77-star penalty across 8,100+ premature reviews	High (Immediate)
2. Merchant SLAs	48h handoff cap; search demotion for breach rate $> 10\%$	Addresses 63.39% of dispatch breaches caused by 5% of sellers	High (Policy)
3. Dynamic Route SLAs	85th-percentile transit model; line-hauls for SP $\rightarrow$ North	Prevents delays exceeding 7 days; curbs 1.8-star churn floor	Medium (Software)
4. Freight De-Stacking	50% marginal shipping discount on additional units	Returns R$ 96k to buyers (-29.5% fees); lifts basket size	High (Immediate)
5. Split-Cart Mitigation	Transparent split-shipment alerts; cross-dock consolidation	Reclaims rating drop from 2.86 to 4.17 stars on multi-seller carts	Medium (UX / 3PL)
6. Payment Optimization	Real-time rails (PIX); $10\times$–$12\times$ installment tiers	Eliminates 29h Boleto delay; unlocks R$ 450+ basket values	High (Fintech)
7. Merchant Decentralization	Regional onboarding hubs across South, Northeast & Central-West	Leverages 1.78-day transit and 4.26-star local delivery advantages	Medium (Strategic)

6 Constraints & Future Improvements

While this empirical investigation leverages over 100,000 commercial transactions to uncover the drivers of e-commerce delivery performance and customer satisfaction, several methodological constraints, operational blind spots, and dataset limitations contextualize our findings. This section details these analytical boundaries and outlines concrete future enhancements for research and engineering deployment.

6.1 Analytical Constraints & Data Limitations

1. **Censorship of Incomplete Orders and Attrition Bias:** Approximately 3,000 orders ($\approx 3\%$) lack a recorded customer delivery timestamp because they were canceled, lost in transit, or actively moving through courier networks when the database snapshot was extracted. Because delivery performance and satisfaction analyses are conditioned strictly upon completed drop-offs, the experiences of the most acutely frustrated consumers are systematically excluded. Additionally, sporadic omissions in intermediate timestamps (e.g., payment approval or carrier handoff) prevent granular root-cause decomposition for every single delayed shipment.

2. **Qualitative Sparsity and Detractor Comment Skew:** While 1-to-5 numerical star ratings are universally recorded, 88.3% of review titles and 58.7% of customer comment bodies were left blank. This creates significant qualitative sparsity for text-mining pipelines. Furthermore, consumers who submit text feedback are over four times more likely to be dissatisfied detractors than silent reviewers; consequently, qualitative NLP analysis skews almost exclusively toward transit failures, leaving the specific drivers of positive sentiment under-documented.

3. **Right-Censorship in Seller Acquisition Cohorts:** Lead conversion metrics suffer from right-censorship for inbound marketing leads generated during April and May 2018. Because closing a prospective merchant requires an average of two weeks and can extend to four or five months for complex enterprise sellers, these recent cohorts exhibit artificially depressed conversion rates solely because their commercial negotiation pipelines remained open at dataset extraction.

4. **Warehouse Operational Blind Spots:** The database records carrier custody transfer timestamps but offers zero visibility into merchant warehouse inventory levels, picking and packing labor capacity, or upstream supplier backorders. Consequently, while chronic SLA breachers are readily identified, it is impossible to determine whether dispatch latency originates from physical stockouts, internal fulfillment bottlenecks, or carrier pickup failures.

5. **Absence of Buyer Demographic Attributes:** Transaction logs exclude demographic dimensions such as consumer age, household income, credit score, or checkout device telemetry. This prevents rigorous causal attribution regarding whether the heavy reliance on Boleto bank slips and multi-installment financing is driven by structural unbanking, generational habits, or pure transaction convenience.

6. **Centroid Approximations and Geodesic Distance Bias:** Raw geolocation records contain over one million redundant GPS coordinates, necessitating aggregation to postal-code prefix medians. Furthermore, computing geographic separation via straight-line geodesic distance ignores topography, highway infrastructure, and courier sorting hubs, understating actual road transit mileage and duration.

7. **Catalog Metadata Imputation:** Exactly 610 product listings lacked category classifications, descriptive text, and photographic dimensions. Imputing these missing physical

measurements and rectifying 0-gram records using category medians introduces synthetic uniformity that may mask physical package irregularities.

8. **Observation Horizon vs. Long-Cycle Customer Churn:** The 20-month continuous window (January 2017 to August 2018) captures a 97% single-order customer share. However, because the marketplace catalog features durable goods (e.g., auto parts, home appliances, furniture) characterized by multi-year replacement horizons, true customer churn cannot be definitively separated from natural inter-purchase intervals within this observation timeframe.

6.2 Future Improvements & Strategic Next Steps

To expand upon this analytical framework and build robust operational production systems, we outline five prioritized engineering and strategic initiatives:

6.2.1 Predictive SLA Engines & Granular Tracking

- **Real-Time Dynamic SLA Modeling:** Replace the uncalibrated 11.2-day static buffer with machine learning models trained on order item count, historical merchant dispatch velocity, route-specific transit distributions, and payment authorization lag. This allows checkout systems to present accurate, lane-calibrated delivery promises rather than arbitrary blanket estimates.
- **Carrier Tracking API Ingestion:** Integrate carrier webhooks to capture intermediate tracking milestones (e.g., regional sorting arrival, customs clearance, out-for-delivery scans) to isolate precisely where packages stall across the 9.3-day road transit phase, especially on long-haul lanes like São Paulo to Pará (17.0 days) and Maranhão (16.4 days).

6.2.2 Sentiment Intelligence & Delivery-Gated Feedback

- **Delivery-Gated Review Prompts:** Hard-code survey dispatches to fire exclusively upon carrier delivery validation, preventing reviews while parcels remain in transit and immediately recovering the unearned 1.5-star penalty affecting 8.4% of delivered orders.
- **Domain-Specific Transformer NLP:** Transition from basic keyword matching (*não*, *recebi*, *atraso*) to pre-trained Portuguese transformer models (e.g., BERTimbau) capable of multi-label aspect classification, sentiment nuance extraction, and automated routing of severe complaints to logistics escalation teams.

6.2.3 Controlled Experimentation on Pricing & Financing

- **A/B Testing Multi-Unit Freight Discounts:** Execute randomized controlled trials testing a 50% incremental shipping discount on multi-quantity items from the same seller to validate whether returning R$ 96,000 in shipping relief lifts basket conversion without degrading merchant margins.
- **Extended Installment Financing Pilots:** Because basket value scales from R$ 90–120 (single payment) to over R$ 450–500 (10 installments), pilot 18× and 24× financing with banking partners to evaluate gross merchandise value elasticity on high-ticket durable categories.

6.2.4 Supply Decentralization & Merchant Governance

- **Regional Merchant Acquisition:** Diversify seller acquisition beyond São Paulo (which currently holds 59.74% of sellers) into the South, Northeast, and Central-West to expand intra-city fulfillment (1.78 days median transit, 4.26 stars, 11.15% detractors) and compress interstate shipping fees.
- **Automated SLA Dispatch Governance:** Enforce automated fulfillment policies on the 5% of chronic breachers causing 63.39% of dispatch delays by down-ranking catalog listings below 90% compliance, while rewarding the 1,600+ zero-breach sellers with buy-box priority.

6.2.5 Payment Streamlining & Multi-Seller Cart UX

- **Promote Instant Payment Rails (PIX):** Introduce targeted checkout incentives for instant digital rails to bypass the 29.07-hour Boleto authorization latency, curbing the 8.90% late-delivery rate stemming from fulfillment holds.
- **Split-Cart Advisories & Cross-Dock Pilots:** Implement prominent checkout warnings for multi-seller orders and pilot regional cross-dock consolidation to protect customer satisfaction from plunging from 4.17 stars to 2.86 stars.

7 Individual Contributions & Work Log

7.1 Individual Contribution Matrix

To ensure academic integrity and transparent evaluation, Table 6 summarizes the individual datasets handled, key project outputs, and recorded effort based directly on team work logs.

Table 6: Individual Contribution and Dataset Summary

Member	Datasets Handled	Key Outputs	Hours
Sanjoli	Customers, Geolocation, Orders, Order Reviews, Order Items, Order Payments, Products, Sellers, Translations	Cleaned and analysed dataset, satisfaction distributions, monthly trends, negative-review themes, delivery-status comparison, Tableau dashboard visuals	34
Jay	Customers, Geolocation, Orders, Order Reviews	Cleaned master data file, customer and delivery distributions, core analytical report sections	32
Joyce	Sellers, Products, Translations, Orders, Order Items, Order Payments, Order Reviews, Customers, Geolocation	Cleaned product catalog, standardized seller dataset, IQR outlier audit, master Google Colab notebook, technical report sections	40
Kishar	Orders, Order Payments	Cleaned master dataset, 12 core EDA charts, multivariate visualizations, executive presentation slide deck	28
Vishal	Order Items, Products, Sellers, Orders, Geolocation, Payments, Reviews, Marketing Funnel	Cleaned line-item records, freight compounding analysis, seller SLA breach metrics, Haversine transit models, strategic recommendations, LaTeX report compilation	36

7.2 Weekly Progress & Work Log

Table 7 details the chronological weekly milestones, technical contributions, and analytical tasks completed by each team member across the three-week project duration.

Table 7: Weekly Progress and Individual Work Log

Member	Week 1	Week 2	Week 3
Sanjoli	Explored review dataset, understood variables, checked missing values and duplicates, and calculated review-score and satisfaction statistics.	Cleaned and preprocessed data, created satisfaction cohorts, performed monthly trend analysis, and identified complaint themes.	Prepared project findings, business recommendations, limitations, and report sections; created Tableau dashboard visuals comparing satisfaction and delivery.
Jay	Cleaned location and customer data, removed duplicate zip codes, and plotted initial charts showing customer distribution.	Combined customer data with orders and reviews. Calculated delivery times and delays, then plotted charts to evaluate review score impacts.	Wrote main sections of the final project report, including summary, data cleaning steps, findings, and business recommendations.
Joyce	Ingested raw datasets, validated data types and nulls, ran IQR outlier analysis on catalog dimensions, and verified seller postal codes.	Standardized catalog schema, mapped English translations, imputed missing dimensions using medians, and plotted seller Pareto charts.	Consolidated all team code pipelines into master Google Colab notebook, and authored core technical report sections.
Kishar	Ingested orders and payments datasets, parsed datetime fields, audited missing values, cleaned anomalies, and merged tables into a cleaned master dataset.	Engineered delivery and payment metrics, and conducted univariate and bivariate exploratory data analysis across volume growth, status, delays, and payments.	Conducted multivariate statistical analysis (heatmaps, radar, pairplot), synthesized key business findings, and developed the final executive presentation slide deck.
Vishal	Audited 112,650 order items and dimension schemas, checked composite keys and missing values, and evaluated IQR price and freight outlier distributions.	Analyzed linear freight compounding, evaluated multi-seller split-cart rating drops, modeled Haversine transit distances, and quantified seller dispatch SLA breach rates.	Synthesized operational solutions, standardized technical report figures and tables, and led LaTeX report compilation across core analytical sections.

8 Technical Appendix & Code Repository

8.1 Relational Data Schema & Entity-Relationship Diagram

The dataset is structured across nine relational tables centered on customer orders, capturing transaction timestamps, catalog dimensions, payment installments, seller fulfillment, and post-purchase review surveys. Figure 9 illustrates the complete relational entity-relationship (ER) diagram, primary keys, and foreign key linkages across the marketplace schema.

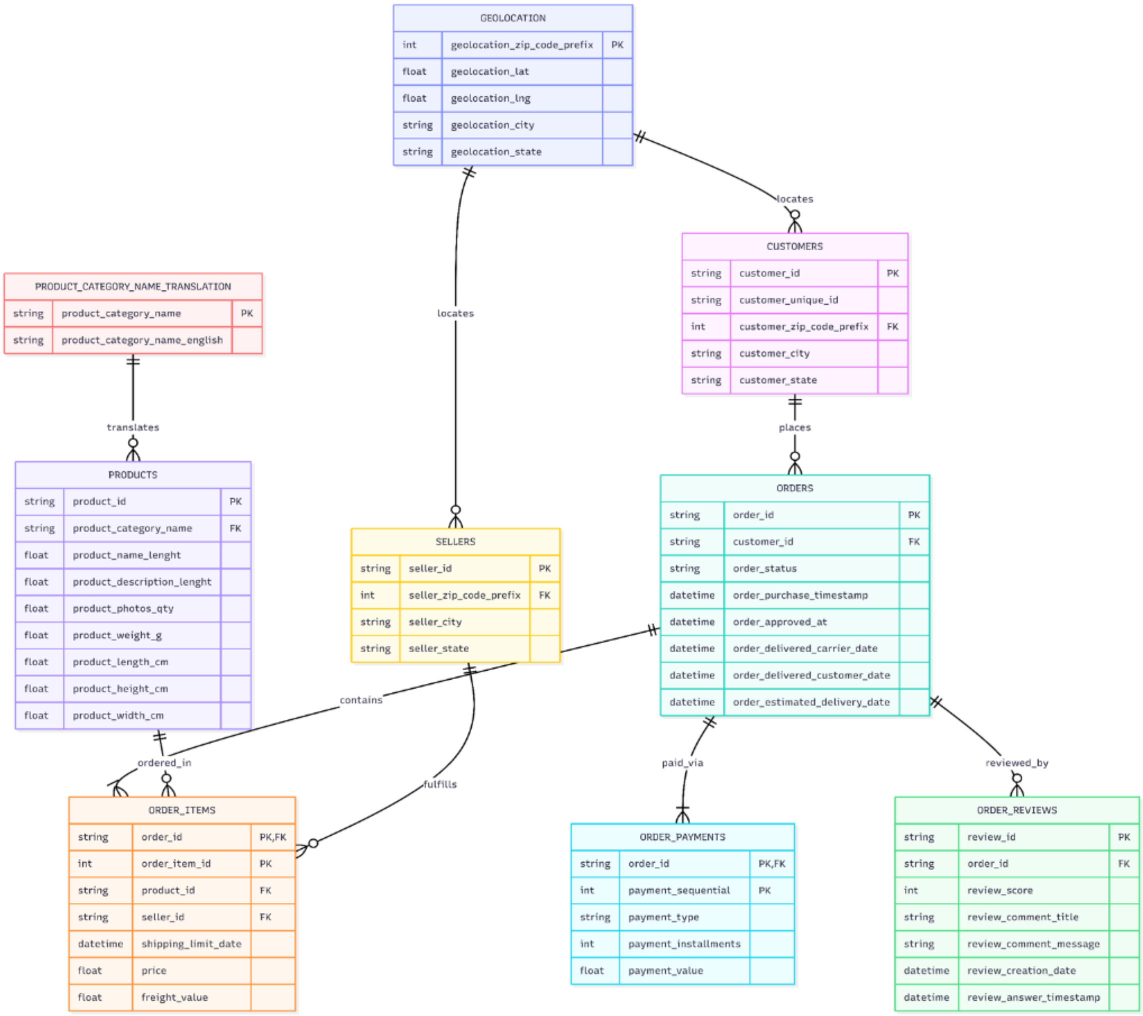

Figure 9: Relational Entity-Relationship (ER) Diagram of the E-Commerce Marketplace Database.

8.2 Operational Metric Definitions

The core operational latencies, delivery deltas, and customer satisfaction indicators engineered across the analysis workflows and evaluated throughout this report are defined as follows:

- **Actual Delivery Duration (`actual_delivery_days`):** Elapsed time in days between customer order placement and verified doorstep delivery:

$$\text{actual\_delivery\_days} = \frac{t_{\text{order\_delivered\_customer\_date}} - t_{\text{order\_purchase\_timestamp}}}{86,400}$$

- **Promised Delivery Delay (`delay_days`):** Variance in days between actual customer arrival and the platform's promised deadline:

$$\text{delay\_days} = \frac{t_{\text{order\_delivered\_customer\_date}} - t_{\text{order\_estimated\_delivery\_date}}}{86,400}$$

- **Late Delivery Indicator (`is_late`):** Binary flag denoting whether an order breached the promised arrival date:

$$\text{is\_late} = \mathbb{I}(\text{delay\_days} > 0)$$

- **Merchant Dispatch Handling (`seller_handling_days`):** Operational window in days taken by a merchant to package and hand over goods to the carrier:

$$\text{seller\_handling\_days} = \frac{t_{\text{order\_delivered\_carrier\_date}} - t_{\text{order\_approved\_at}}}{86,400}$$

- **Carrier Transit Duration (`carrier_transit_days`):** Road transit duration in days between carrier custody handoff and customer delivery:

$$\text{carrier\_transit\_days} = \frac{t_{\text{order\_delivered\_customer\_date}} - t_{\text{order\_delivered\_carrier\_date}}}{86,400}$$

- **Customer Dissatisfaction Indicator (`is_dissatisfied`):** Binary flag capturing low review ratings (1 or 2 stars):

$$\text{is\_dissatisfied} = \mathbb{I}(\text{review\_score} \in \{1, 2\})$$

8.3 Code Repository & Software Packages

All analysis scripts, data preprocessing workflows, and interactive dashboard files are maintained in the following resources:

- **Dataset Source (Google Drive):** https://drive.google.com/drive/folders/1yKVQgc-7WJsHrDhyyC3kObWn627tQvBG?usp=drive_link
- **Executable Analysis Notebook (Google Colab):** https://colab.research.google.com/drive/1qBzxqBBF6WzjzhQiGXoTcUlhBdwRydt8?usp=sharing
- **Software Packages & Libraries:** `pandas`, `numpy`, `matplotlib`, `seaborn`, `streamlit`, `os`, `re`, `warnings`, `collections`.